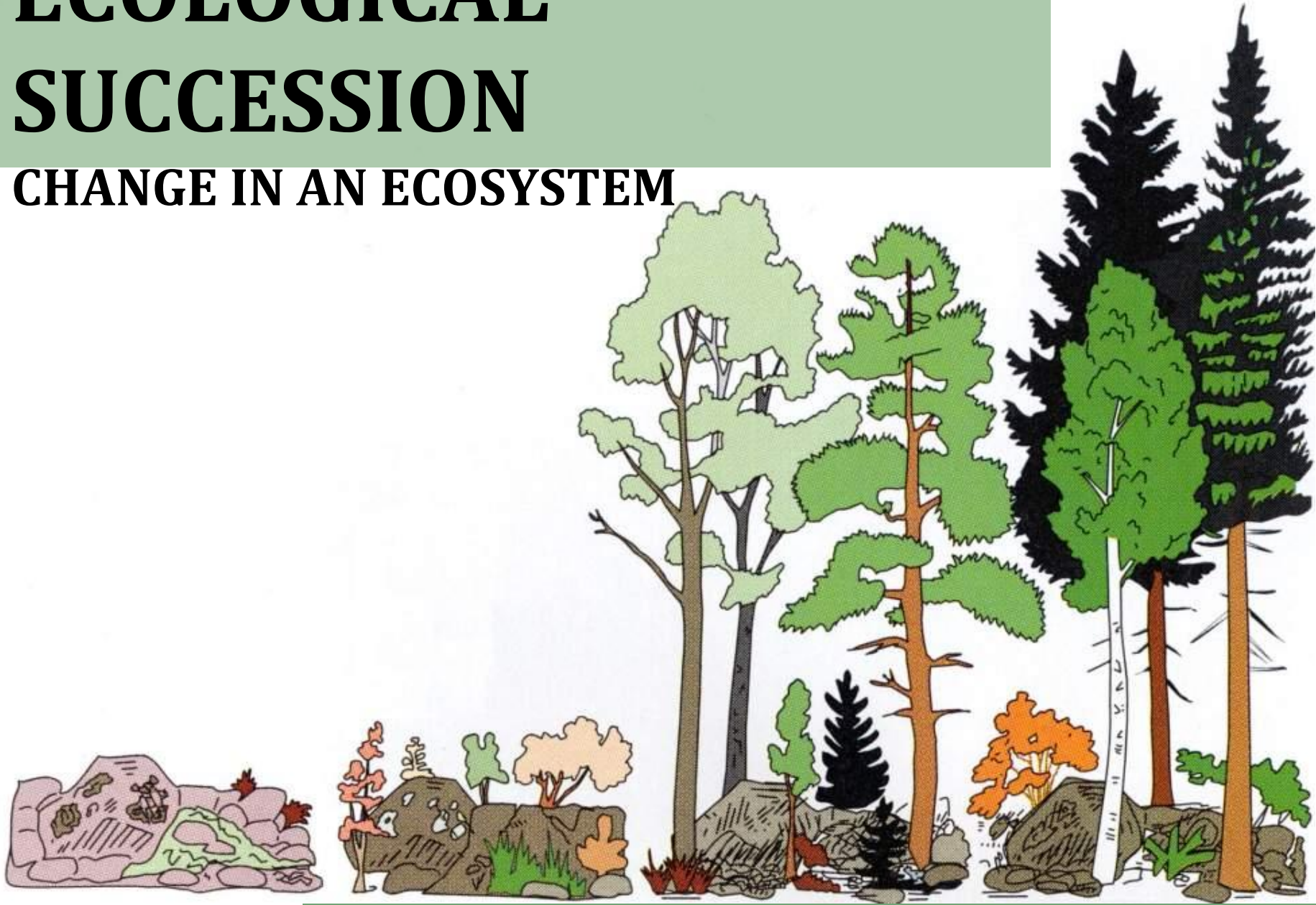


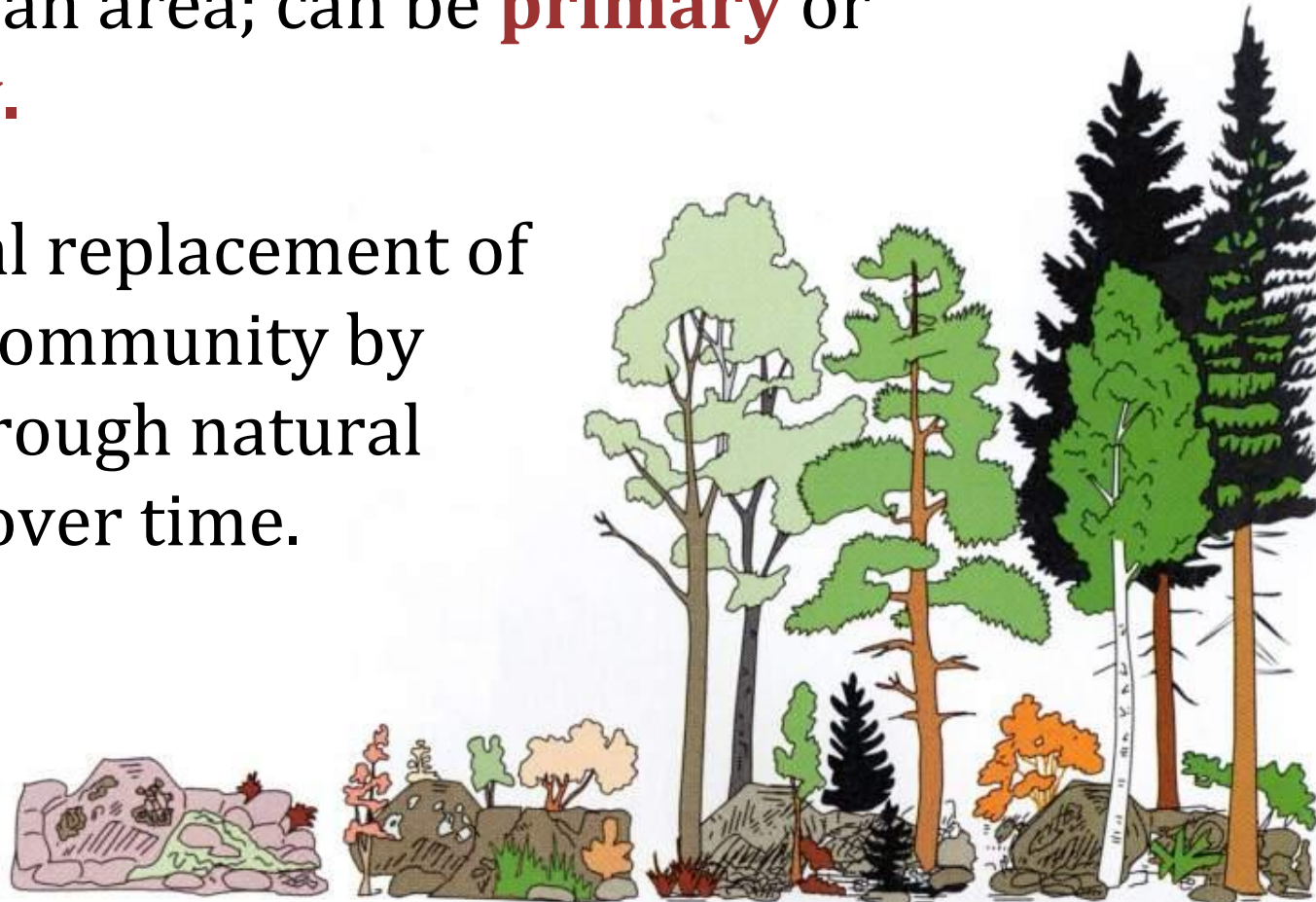
ECOLOGICAL SUCCESSION

CHANGE IN AN ECOSYSTEM



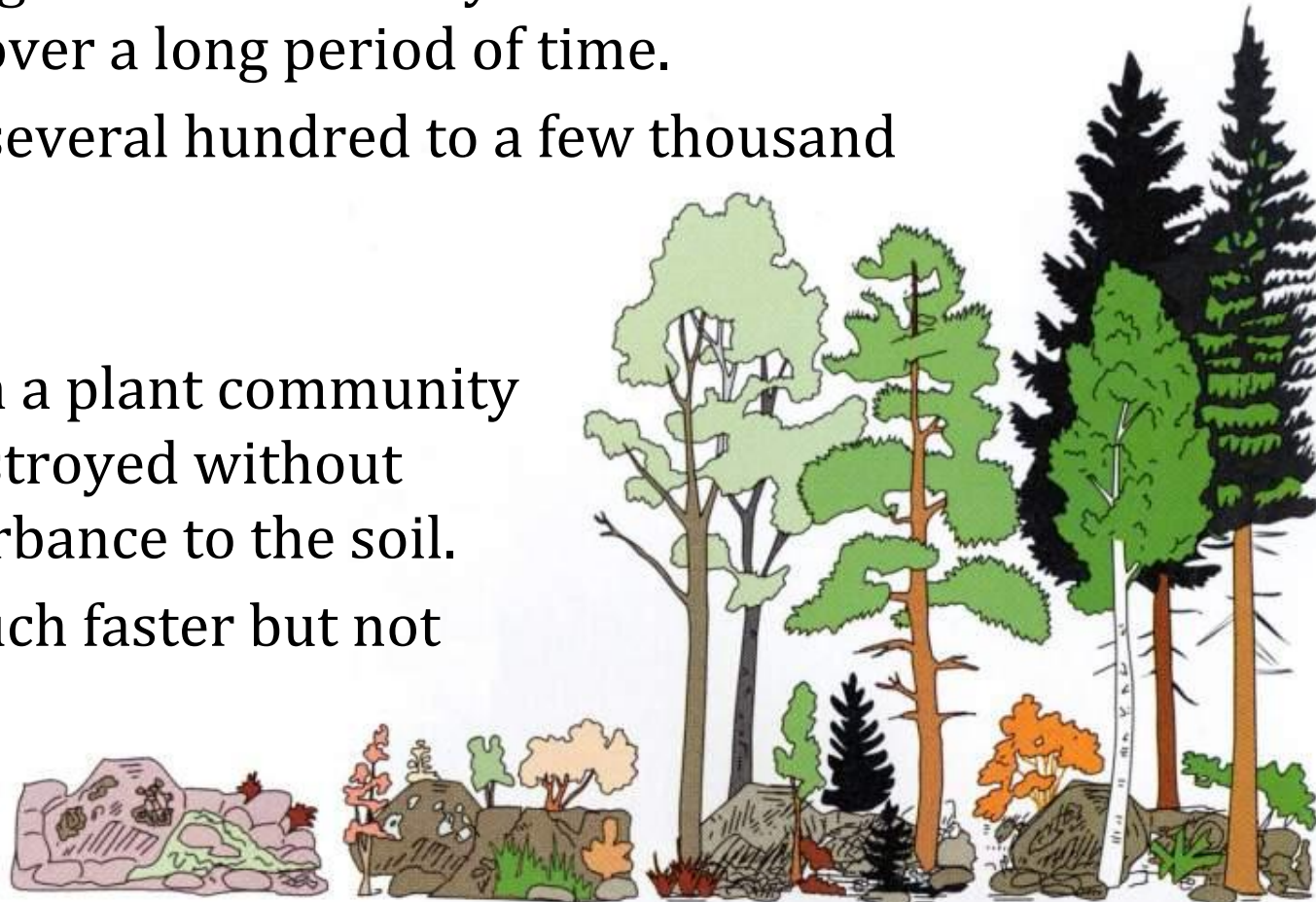
What is Succession?

- Natural, gradual changes in the types of species that live in an area; can be **primary** or **secondary**.
- The gradual replacement of one plant community by another through natural processes over time.



Primary vs. Secondary Succession

- Primary
 - The gradual growth of an ecosystem in an area lacking soil over a long period of time.
 - Often takes several hundred to a few thousand years.
- Secondary:
 - Occurs when a plant community has been destroyed without severe disturbance to the soil.
 - Typically much faster but not always.



Primary Succession

- Begins in a place without any soil:
 - Sides of volcanoes
 - Landslides
 - Flooding
- Starts with the arrival of living things such as lichens that do not need soil to survive.
 - These organisms like lichens are called **PIONEER SPECIES**.



Primary Succession

- Soil starts to form as lichens and the forces of weathering and erosion help break down rocks into smaller pieces and move them to new areas.
- When lichens die, they decompose, adding small amounts of organic matter to the rock to make soil.



Primary Succession

- Simple plants like mosses and ferns can grow in the new soil.



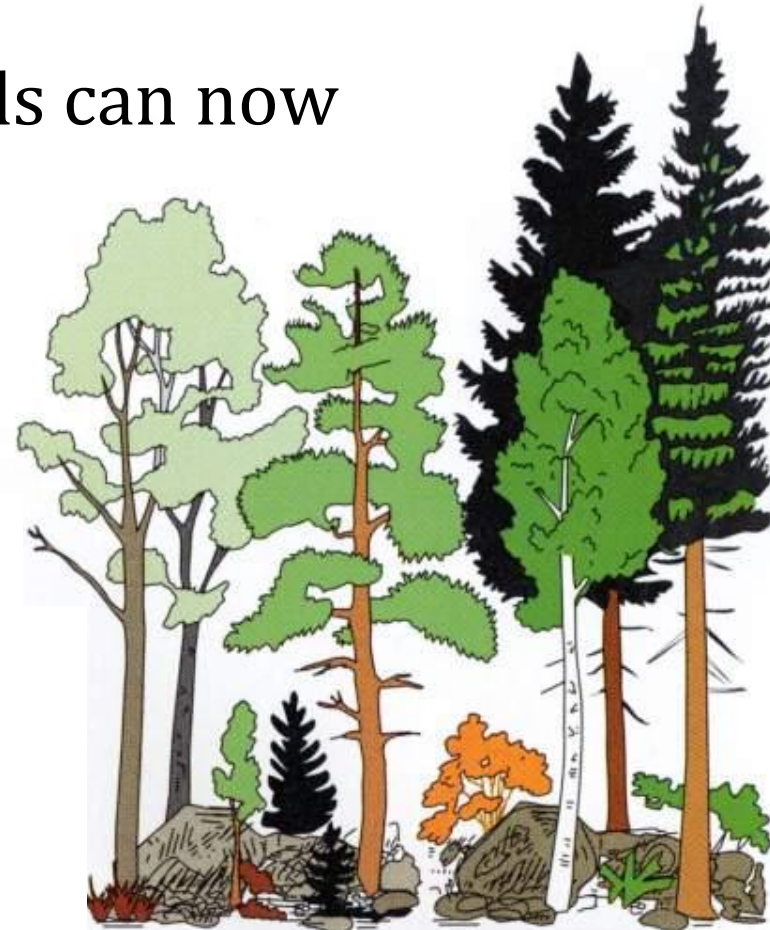
Primary Succession

- ❑ The simple plants die, adding more organic material to the soil.
- ❑ The soil layer thickens. Grasses, wildflowers, other plants, and small animals begin to take over.

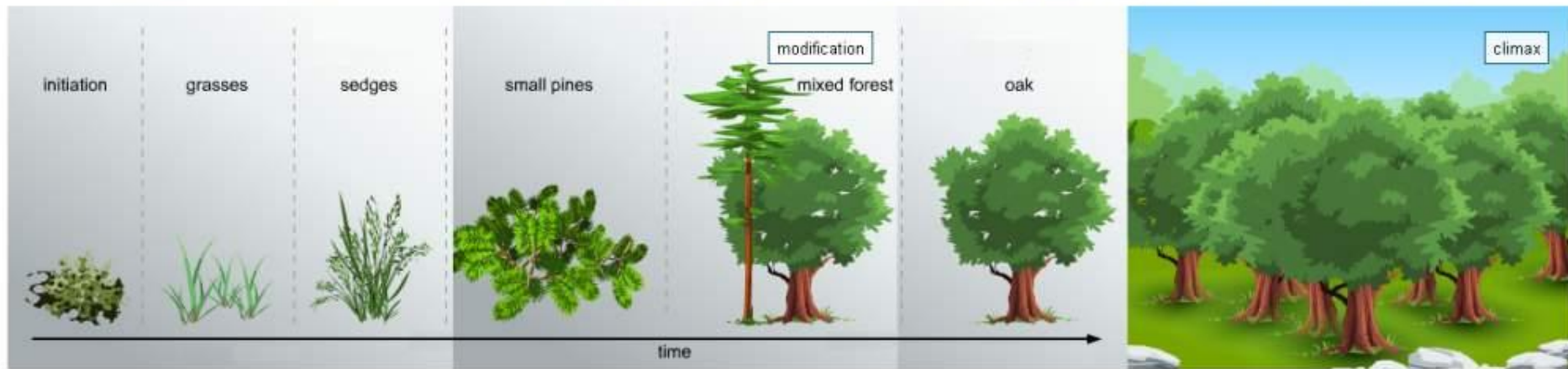


Primary Succession

- As grasses and other plants die, they add more nutrients to the soil.
- Shrubs, trees and larger animals can now survive in the area.

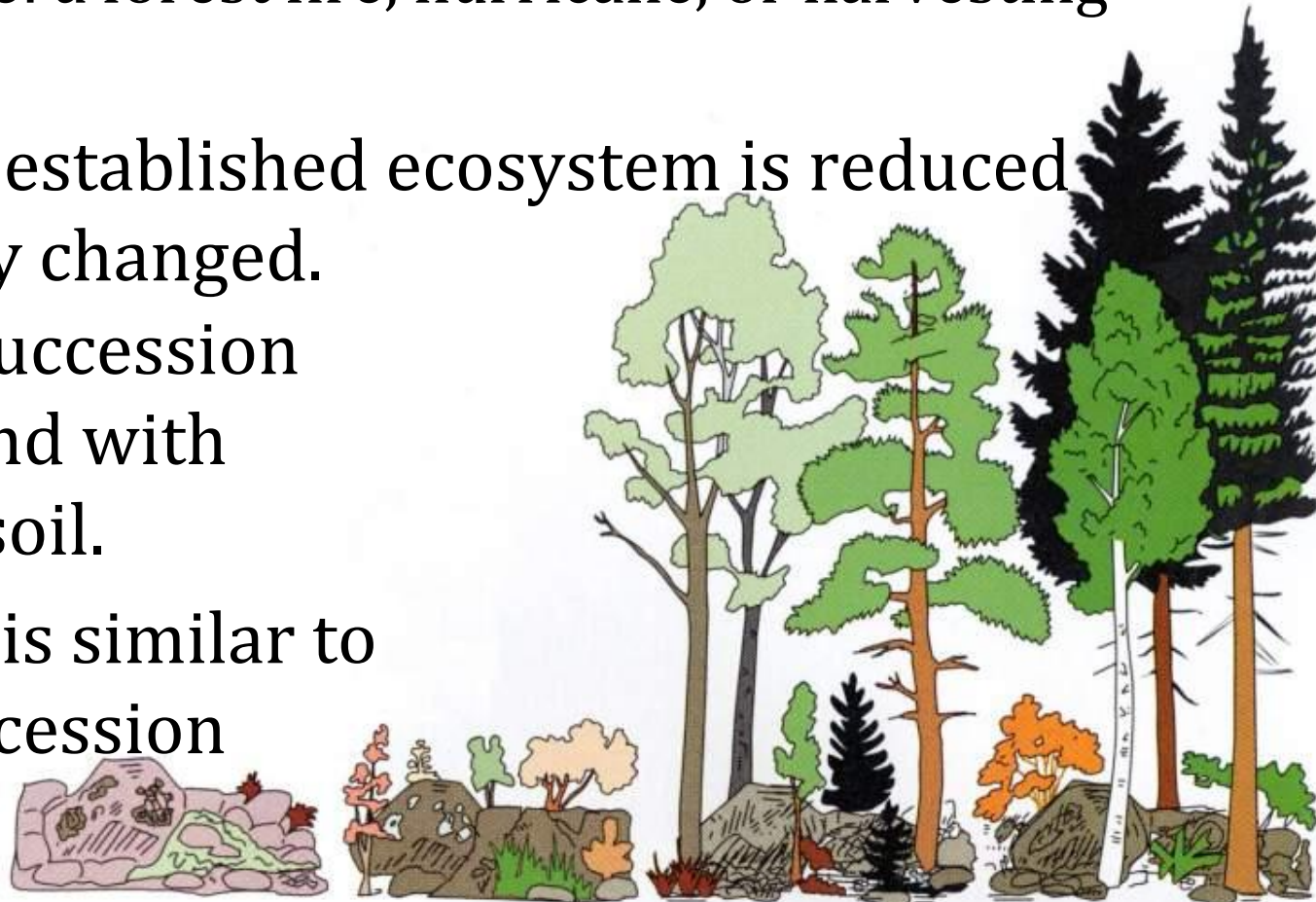


Primary Succession



Secondary Succession

- Secondary Succession is initiated by an event.
 - For example: a forest fire, hurricane, or harvesting a crop
- The already-established ecosystem is reduced or drastically changed.
- Secondary Succession occurs on land with preexisting soil.
- The process is similar to Primary Succession thereafter.



Secondary Succession

